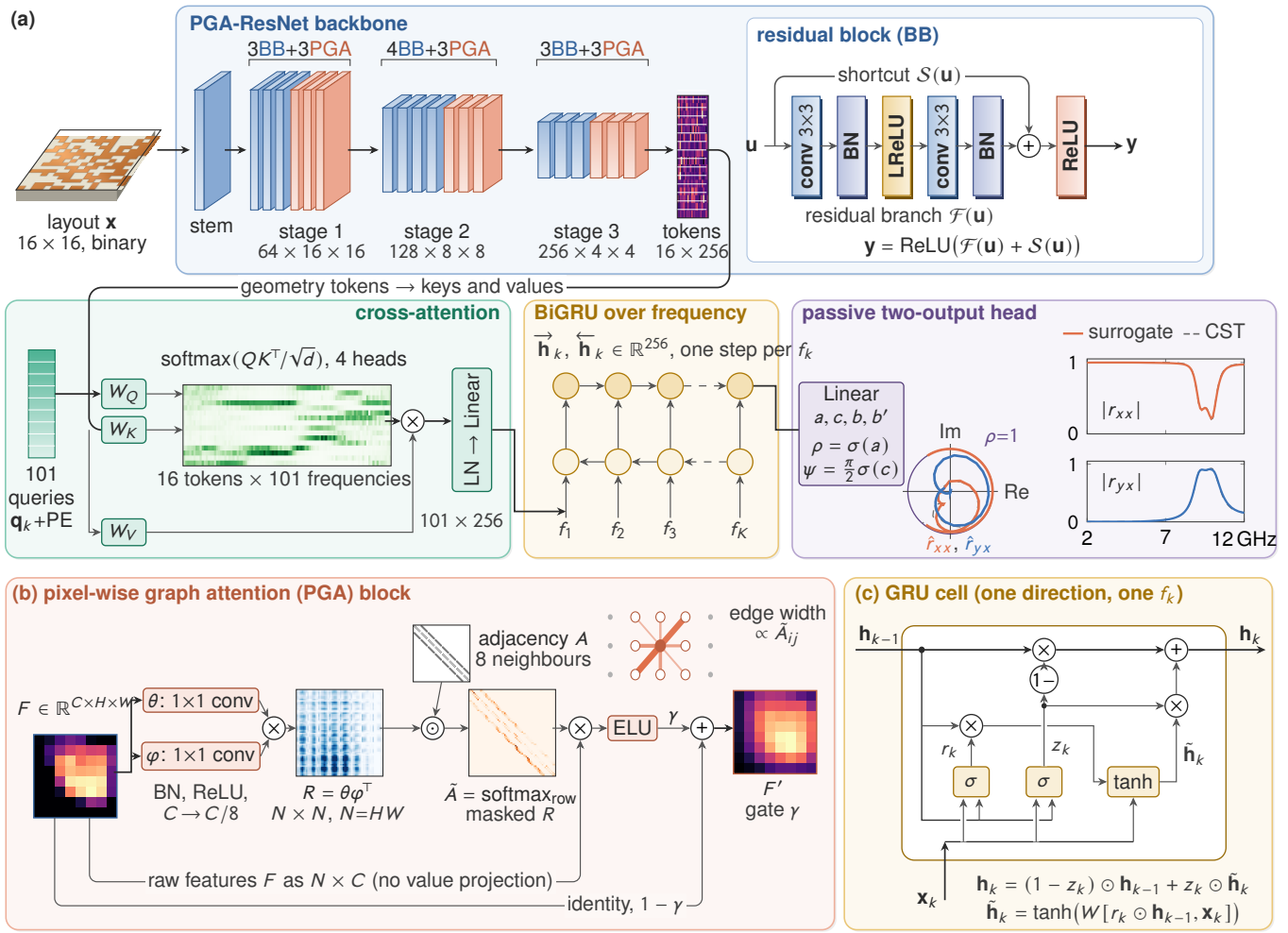

(a)
PGA-ResNet backbone
3BB+3PGA
4BB+3PGA
3BB+3PGA
layout x
16 × 16, binary
stem
stage 1
64 × 16 × 16
stage 2
128 × 8 × 8
stage 3
256 × 4 × 4
tokens
16 × 256
residual block (BB)
shortcut $\mathcal{S}(\mathbf{u})$
u
conv 3×3
BN
LReLU
conv 3×3
BN
ReLU
y
residual branch $\mathcal{F}(\mathbf{u})$
$\mathbf{y} = \mathrm{ReLU}(\mathcal{F}(\mathbf{u}) + \mathcal{S}(\mathbf{u}))$
geometry tokens → keys and values
cross-attention
softmax$(QK^\top/\sqrt{d})$, 4 heads
101
queries
$\mathbf{q}_k$+PE
W_Q
W_K
W_V
16 tokens × 101 frequencies
LN → Linear
101 × 256
BiGRU over frequency
$\overrightarrow{\mathbf{h}}_k$, $\overleftarrow{\mathbf{h}}_k \in \mathbb{R}^{256}$, one step per f_k
f_1 f_2 f_3 f_K
passive two-output head
Linear
a, c, b, b'
$\rho = \sigma(a)$
$\psi = \frac{\pi}{2}\sigma(c)$
Im
Re
ρ=1
$\hat{r}_{xx}$, $\hat{r}_{yx}$
surrogate -- CST
$|r_{xx}|$
$|r_{yx}|$
1
0
2
7
12 GHz
(b) pixel-wise graph attention (PGA) block
$F \in \mathbb{R}^{C\times H\times W}$
θ: 1×1 conv
φ: 1×1 conv
BN, ReLU,
$C \to C/8$
$R = \theta\varphi^\top$
$N \times N$, $N=HW$
adjacency A
8 neighbours
edge width
$\propto \tilde{A}_{ij}$
$\tilde{A} = \mathrm{softmax}_{\mathrm{row}}$
masked R
ELU
γ
F'
gate γ
raw features F as $N \times C$ (no value projection)
identity, $1 - \gamma$
(c) GRU cell (one direction, one f_k)
$\mathbf{h}_{k-1}$
$\mathbf{h}_k$
1−
r_k
z_k
$\tilde{\mathbf{h}}_k$
σ
σ
tanh
$\mathbf{x}_k$
$\mathbf{h}_k = (1 - z_k) \odot \mathbf{h}_{k-1} + z_k \odot \tilde{\mathbf{h}}_k$
$\tilde{\mathbf{h}}_k = \tanh(W[r_k \odot \mathbf{h}_{k-1}, \mathbf{x}_k])$